ESTR1002 Fall 2025

Lab Exercise #6

Due: 22 Oct 2025

Separate your answer for each question in a different .c file. Compress all files into a .zip and submit it to Blackboard.

Question 1

A strong password must contain all the following elements:

- punctuation, including . ; : ( ) < > {} [ ] " ? ~ ! ` ' & % # | \ /
- both upper and lower cases, and
- a number.

Input: a string of length in the range of [6, 30].

Output: if the input is a strong password, output “yes”, otherwise, output “no”.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
password
no
```

Sample Run #2:

```
PASSword123
no
```

Sample Run #3:

```
P"eNm\ga7=W;VQm
yes
```

Question 2

Given a string s composed of only alphabets and symbols (without digits), you need to compress repeating characters to a format of [character + repeating time], where the repeating time should be greater than or equal to 2.

Input: a string s composed of only alphabets and symbols (without digits), in which the total number of characters in s is no greater than 2000.

Output: the compressed string.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
aaackhhGGGg
a3ckh2G3g
```

Sample Run #2:

```
AbcdC#$
AbcdC#$
```

Question 3

An anagram is a type of word play, rearranging the letters of a word or phrase to produce a new word or phrase, using all the original letters exactly once; for example **Torchwood** can be rearranged into **DoctorWho**.

Let us determine if two strings are a pair of anagrams in a case-insensitive way. On top of that, we would not take space characters into account. That means your program should not consider any space characters while it is reading the input.

Input: Two strings on two lines. Each string only contains upper-case, lower-case, and space characters only. The length of each string ranges from 1 to 1,000. Note that each string should have at least one upper-case / lower-case character.

Output: If the first string is an anagram of the second string, print "yes". Else, print "no".

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
Torch Wood
Doctor Who
yes
```

Sample Run #2:

```
TOM MARVOLO RIDDLE
I AM LORD VOLDEMORT
yes
```

Sample Run #3:

```
OK YOU WIN
USE WINDOW
no
```

Question 4

You are given a string consisting of only English letters (both uppercase and lowercase) and an integer K. Your task is to shift only the vowels in the string by K positions, while leaving all other characters in their original positions. The vowels in the English alphabet are "a, e, i, o, u".

Rules:

- Leftward Shift: If K is a positive number, shift the vowels to the left by K positions.
- Rightward Shift: If K is a negative number, shift the vowels to the right by - K positions.
- Circular Shifting: The shifting is circular, meaning that after shifting past the beginning or end of the vowel sequence, the vowels wrap around. For example, shifting the first vowel to the left would move it to the last position in the sequence.
- Case Preservation: The case (uppercase or lowercase) of each vowel must be preserved after shifting.

Input: there are two lines of inputs. The first line is a string s, in which the number of characters is no larger than 200. All the characters are in the standard 7-bit ASCII table. The second line is the number K.

Output: the processed string where all the vowels are shifted.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
hEllo
1
hollE
```

Sample Run #2:

```
aeiou
2
iouae
```

Sample Run #3:

```
codetEsthAha
-2
cAdatosthehE
```

Question 5

Aliens come to Earth again. But, this time, they are no longer friendly. They send our scientist a message in order to tell us the number of seconds before the war against us begins.

Unfortunately, we have no idea what the message means. We have decided that each symbol indicates one digit, but we are not sure what each digit means or what base the aliens are using. For example, if they wrote "ab2ac999", they could have meant

- "31536000" in base 10, exactly one year, or
- they could have meant "12314555" in base 6, 398951 seconds, or about four and a half days.

We are sure of four things:

1. the number is positive;
2. the aliens will never start a number with a zero;
3. they read a number from left to right, i.e., the leftmost symbol is the most significant value;
4. they are not using unary (base 1).

Your job is to determine the **minimum possible number of seconds** before war begins.

Input: One string of length ranging from one character to 60 characters. The string only contains lower-case characters and digits, without any space characters or punctuations.

Output: The minimum number of seconds before war begins. You can assume that the output is not exceeding 10^{18}.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
11001001
201
```

Sample Run #2:

```
cats
75
```

Sample Run #3:

```
zig
11
```